

Computer Organization & Architecture (CSC-201), Autumn 2026

Assignment: Processor Design

Team Size

Min 4 and Max 5 (CRs need to coordinate and prepare an Excel Sheet for different groups and resolve conflicts. TAs will help manage the business. Share the final Excel sheet with me.)

Problem Statement

1. Design
 - a. a 4-stage and 6-stage RISC201 ISA. Compare the behavior of both processors.
 - b. a corresponding microprogrammed control unit (you are free to choose a horizontal or vertical model).
 - c. A corresponding ALU with efficient arithmetic and logical operations.
2. Implement a complete software environment and simulation framework. The software suite must encompass
 - a. A two-pass assembler that translates RISC201 assembly programs into binary or hexadecimal representations.
 - b. a disassembler that parses binary or hexadecimal word strings and reconstructs assembly instructions.
 - c. an interactive command-line interface to simulate register file, instruction execution pipeline,
 - d. a step-by-step CPU execution simulator with debugger capabilities, and an interactive CLI debugger.
 - e. stack preprocessor macro expansion to incorporate preprocessor support for full-descending stack operations.
 - f. stack boundary safety verification, and a runtime exception framework to incorporate runtime memory protection against stack corruption.
 - g. An active stack memory visualizer to implement an ASCII stack memory frame renderer.
3. Add the microprogrammed control logic directly to your CPU simulator to execute RISC201 using Control Memory microcode rather than hardwired logic. More appreciation for implementing both horizontal and vertical microprogrammed control units and comparing the behaviour. Make sure to extend the CLI simulator interface to display the microPC's step-by-step state and raise exceptions where necessary.
4. Add an optimized RISC201 ALU class into your simulator code to drive all arithmetic, shift, and logical operations.
5. Evaluate the processor, control unit, and ALU for different possible test cases by considering 4-stage and 6-stage ISAs, different assembly programs, horizontal vs. vertical microprogrammed controllers (optional), and different ALU algorithms.

Timelines

1. *Doubts Clarification Dates:* September 9 and September 16 (During Tut time).
2. *Assignment First Evaluation Dates:* September 30 and October 14, during Tut time and in my office after 6 pm on both dates. A detailed schedule will be prepared by the CRs.
3. *Assignment Final Evaluation Dates:* November 10 and November 11, during Tut time and in my office after 6 pm on both dates. A detailed schedule will be prepared by the CRs.